

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool: Application-Based Questions

Weightage: 40%

QUESTIONS: Total Marks: 30

1: Online Symposium Portal

A college is developing an online symposium registration portal. The following HTML structure and HTTP interaction is observed:

```
<form action="/register" method="POST">
  <input type="text" name="name">
  <input type="email" name="email">
  <button>Submit</button>
</form>
```

When a student submits the form, the browser sends an HTTP request to the server, and the server responds accordingly.

Questions:

1. Identify appropriate **HTML5 semantic elements** missing in the structure.
2. Modify the structure to make it **standards-compliant**.
3. Interpret the **HTTP request generated** during form submission.
4. Analyze the **HTTP response cycle** from server to browser.
5. Suggest improvements for **usability and accessibility**.

2: Cross-Browser Compatibility Issue

A company website shows inconsistent layout across browsers. The following HTML snippet

is used:

```
<html>
  <head>
    <title>Company</title>
  </head>
  <body>
    <div>
      <h1>Welcome
      <p>About us
    </div>
  </body>
</html>
```

Questions:

1. Identify **improper nesting and missing closing tags**.
2. Analyze how different browsers may interpret this structure.
3. Identify **missing semantic elements** (e.g., header, section).
4. Provide a **corrected W3C-compliant HTML structure**.
5. Suggest techniques to ensure **cross-browser compatibility**.

3: Form Submission Failure

A web application fails to send form data to the server. The following configuration is observed:

```
<form action="/submit" method="GET">
  <input type="text" name="username">
  <input type="password" name="password">
  <button type="button">Login</button>
</form>
```

Questions:

1. Identify issues in the **form configuration**.
2. Analyze why the **HTTP request is not triggered**.
3. Interpret the role of **GET method in this scenario**.
4. Propose a **corrected implementation**.
5. Suggest improvements for **secure data transmission**.

Code of Conduct:

I M.RESHWANTH(192411235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:M.RESHWANTH

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

CSA4305-Internet Programming

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Case Study 1: Online Symposium Portal

1. Identify appropriate HTML5 semantic elements missing in the structure.

- `<header>`
- `<main>`
- `<section>`
- `<form>`
- `<footer>`
- `<label>`

2. Modify the structure to make it standards-compliant.

```
<!DOCTYPE html>
<html>
<head>
  <title>Online Symposium Registration</title>
</head>
<body>
<header>
  <h1>Symposium Registration</h1>
</header>
<main>
<section>
```

```
<form action="/register" method="POST">
  <label>Name:</label>
  <input type="text" name="name" required><br><br>
  <label>Email:</label>
  <input type="email" name="email" required><br><br>
  <button type="submit">Submit</button>
</form>
</section>
</main>

<footer>
  <p>&copy; 2026 College Symposium</p>
</footer>

</body>
</html>
```

3. Interpret the HTTP request generated during form submission.

- Browser sends an **HTTP POST** request.
- Form data (name and email) is sent in the request body.
- The request is sent to **/register**.

4. Analyze the HTTP response cycle from server to browser.

- Browser sends the request.
- Server receives and processes it.
- Server stores the data.
- Server returns an HTTP response (200 OK or Redirect).
- Browser displays the response page.

5. Suggest improvements for usability and accessibility.

- Add labels for all fields.
- Use the required attribute.
- Display validation messages.
- Improve keyboard accessibility.
- Use semantic HTML elements.

Case Study 2: Cross-Browser Compatibility Issue

1. Identify improper nesting and missing closing tags.

- Missing `</h1>`
- Missing `</p>`
- Missing semantic elements
- Improper nesting of tags

2. Analyze how different browsers may interpret this structure.

- Chrome may auto-correct the HTML.
- Firefox may render differently.
- Safari and Edge may show layout inconsistencies.

3. Identify missing semantic elements.

- `<header>`
- `<main>`
- `<section>`

- `<footer>`

4. Provide a corrected W3C-compliant HTML structure.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Company</title>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1>Welcome</h1>
```

```
</header>
```

```
<main>
```

```
<section>
```

```
<p>About us</p>
```

```
</section>
```

```
</main>
```

```
<footer>
```

```
<p>Copyright 2026</p>
```

```
</footer>
```

```
</body></html>
```

5. Suggest techniques to ensure cross-browser compatibility.

- Use HTML5 standards.
- Validate code using W3C Validator.
- Use CSS Reset or Normalize.css.
- Test in Chrome, Firefox, Edge, and Safari.
- Use responsive design.

Case Study 3: Form Submission Failure

1. Identify issues in the form configuration.

- Button type is "button" instead of "submit".
- Password should not be sent using GET.

2. Analyze why the HTTP request is not triggered.

- `type="button"` does not submit the form.
- Only `type="submit"` triggers form submission.

3. Interpret the role of GET method in this scenario.

- GET appends data to the URL.
- Password becomes visible in the address bar.
- GET is not secure for login forms.

4. Propose a corrected implementation.


```
<!DOCTYPE html>
<html>
<head>
<title>Login Form</title>
</head>

<body>

<form action="/submit" method="POST">

<label>Username</label>
<input type="text" name="username" required><br><br>

<label>Password</label>
<input type="password" name="password" required><br><br>

<button type="submit">Login</button>

</form>

</body>
</html>
```

5. Suggest improvements for secure data transmission.

- Use **POST** method.
- Use **HTTPS**.
- Validate user input.
- Encrypt passwords before storing.
- Use authentication and session management.